

The background is a light beige color with a fine, grainy texture. In the corners, there are abstract geometric elements: two parallel diagonal lines in the top-left, a circle with a diagonal line through it in the top-right, a single vertical line in the bottom-left, two parallel diagonal lines in the bottom-center, a large circle in the bottom-right, and a single vertical line in the bottom-right corner.

**Are you hungry
and in a hurry to
get to class?**



Meal Vending Machine At USC

Group H:

Anna Natasha, Aaryavardhan Rathore, Donya Damavandi,
Jay Maniyar, Samantha Sy, Yiwen Mei, Yuqing (Helen) Yang

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Problem

At USC getting a healthy, filling meal is **inconvenient**. With only drink machines and cafeterias, the long line and prep time takes forever to get food!

01

Introduction

Food Vending Machine

The perfect solution for busy individuals who want to enjoy **healthy** and **delicious meals** on-the-go in under **3 minutes!**



02

Method: Conjoint Survey

Conjoint Survey with 100 respondents

Should USC Install Food Vending Machines?

1. Understand the demand for a food vending machine.
2. Identify the attributes students and staff care about in meals.
3. Inform feature selection and pricing strategy based on preferences.



Key Attributes



Frozen/Fresh

Frozen (1 month shelf life)

Fresh (3-4 days of shelf life)



Meal Price

\$9, \$12, or \$15



Rotating Menu

Yes: the menu changes every week

No: the menu is fixed



Vegan/Vegetarian Dish

Yes: This is a vegan or vegetarian dish

No: The product is not vegetarian-friendly



Portion Size

Medium (500-700 calories)

Large (700-900 calories)



Behavioral Questions

- Age Group
- Role at USC
- Gender Identity
- Meal Purchase Habits:
How often do you purchase meals at school per week?



03

Analysis: Survey Demographics

- Survey completed by a total of **100 respondents**, with an average response time of **4 minutes**
- Most of our survey respondents are **graduate students aged between 22 and 34**
- 64% of the respondents are male, while 36% are female
- **65% of the respondents purchase meals at school 1-3 times per week**, 13% purchase meals at school more than 3 times per week, and 22% never purchase meals at school

03

Analysis: Willingness-To-Pay

Willingness-To-Pay for an attribute level change is the max \$ amount a customer would pay for the change.

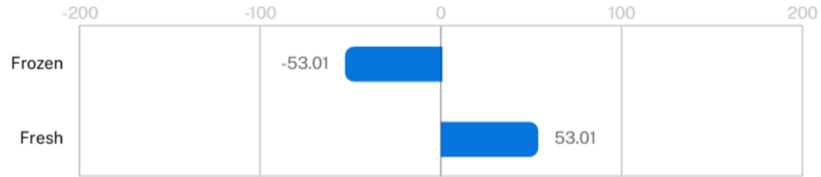
$WTP = \text{Utility difference} * \text{Price per Utility Unit}$

Dollar value of 1 utility = 0.036

- Customers are willing to pay **\$3.87 more** for Fresh over Frozen.
- Customers are willing to pay **\$0.03 more** for a Vegan/Vegetarian Dish.
- Customers are willing to pay **\$1.91 more** for a Rotating Menu.
- Customers are willing to pay **\$0.16 more** for a Large Portion over Medium.

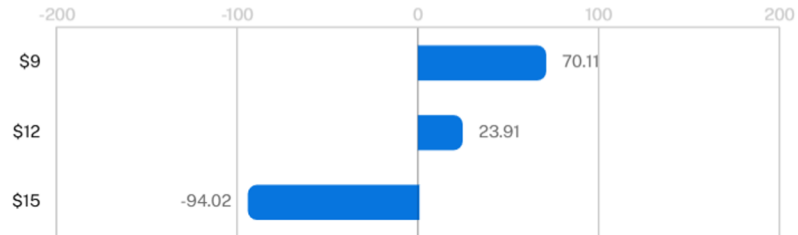
Frozen/Fresh

N=100



Meal Price

N=100



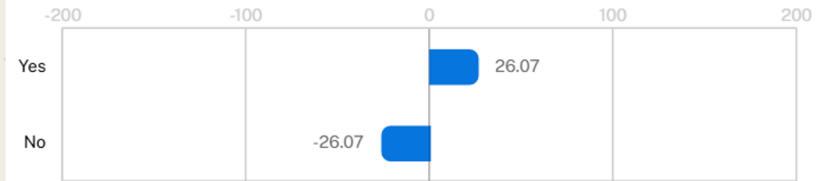
Vegan/ Vegetarian Dish

N=100



Rotating Menu

N=100



Portion Size



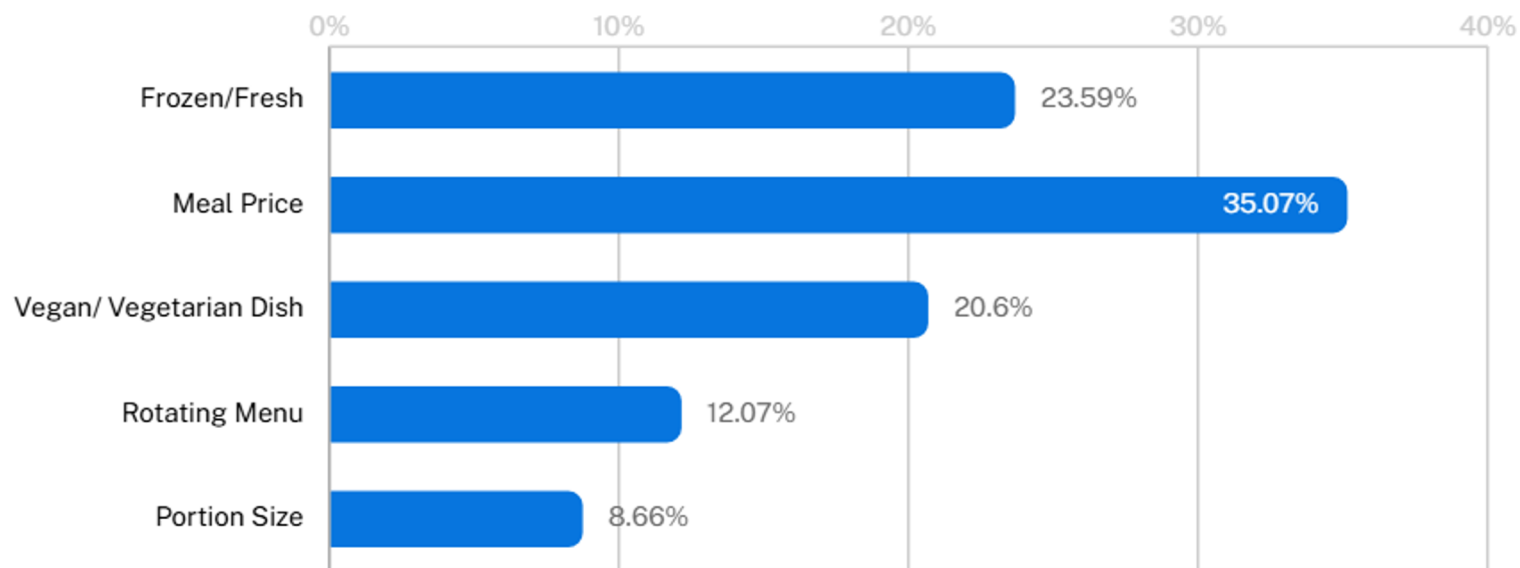
None option

Utility score: -91.46

Relative attribute importance

Attribute importance

N=100





Segmentation Analysis

3 Segment

5 Segment

Groups	AIC	BIC
1	915.96	943.90
2	869.54	929.42
3	832.27	924.08
4	802.58	926.31
5	788.28	943.95

3 Segment Sizes	12.1%	17.2%	70.7%
Attribute Importances			
Frozen/Fresh	8.94	23.32	29.07
Meal Price	66.19	21.73	47.79
Rotating Menu	4.48	0.69	13.17
Vegan/ Vegetarian Dish	14.28	52.59	3.65
Portion Size	6.11	1.67	6.32
None	140.87	66.18	-372.24

5 Segment Sizes	13.3%	27.6%	17.8%	22.6%	18.7%
Attribute Importances					
Frozen/Fresh	6.7	52.8	23.4	14.9	15.7
Meal Price	76.6	13.2	22.6	40.3	71.6
Rotating Menu	3.4	20.6	0.1	12.7	2.2
Vegan/ Vegetarian Dish	8.9	6.3	51.9	5.5	5.2
Portion Size	4.2	6.9	1.4	27.2	5.1
None	144.22	-243.41	63.91	-306.03	-287.12



Customer Profiles



Budget Meal Seekers

12.1% population

Highly price-sensitive customers with utility value that they're **unlikely to consider our product**



Healthy Enthusiasts

17.2% population

Heavy emphasis on fresh, vegan/vegetarian meals and are open to considering our product



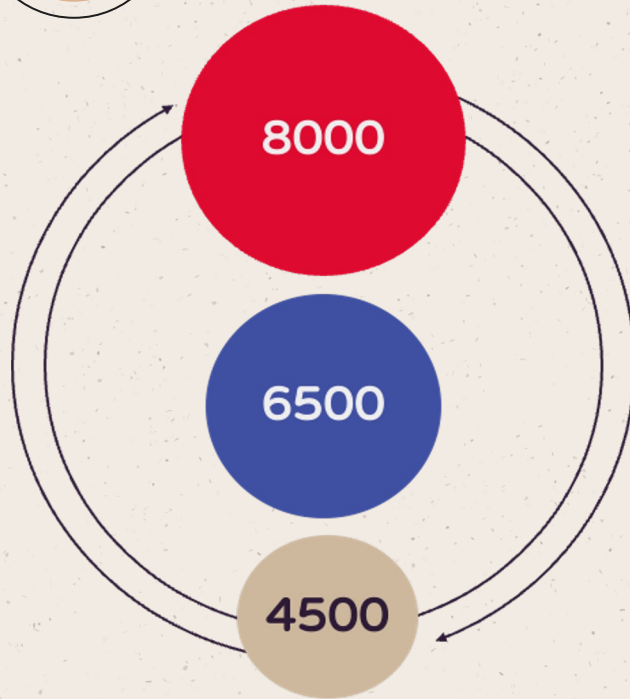
Adventurous Fresh Foodies

70.7% population

Majority of the population who values fresh, good-quality meals and enjoys exploring new dishes that are reasonably priced- **most receptive to our product**

05

Market Sizing Overview



Total Addressable Market (TAM)

Marshall Undergrads (4,200), Grads (3,300),
Faculty & Staff (500)

Serviceable Addressable Market (SAM)

Based on survey results, Marshall Undergrads
(3,600), Grads (2,500), Faculty & Staff (400)
that consume food at Marshall

Serviceable Obtainable Market (SOM)

Target Audience: Marshall Undergrads (2,500),
Grads (1,700), Faculty & Staff (300)

05

Profit Analysis



Revenue

Price: 12\$
WTP ranges: 9-12 and 12-15
Weekly Demand: 13,000
Target Segment: 9,000 (70%)
Adoption: 2,000 (24%)



Costs

Fixed Costs: 15,000\$
Unit Costs (variable): 6\$
Additional Costs: 3\$
Margin: 20.0%



Profits Per Week

Revenue - Costs = **\$2,500**
(24,000-21,500)

Recommendations and Managerial Implications



Pricing Strategy

- Willing to pay **\$3.5 more for fresh food** over frozen meals
- The **seasonal products can be marked at \$2 more** than the regular products

Optimal Implementation Strategy

- Capacity: **500 units/week, Profit: \$600/week**
- Will start with JKP and JFF
- Next, between HOH and BRI
- As we start earning profits, we implement all across USC



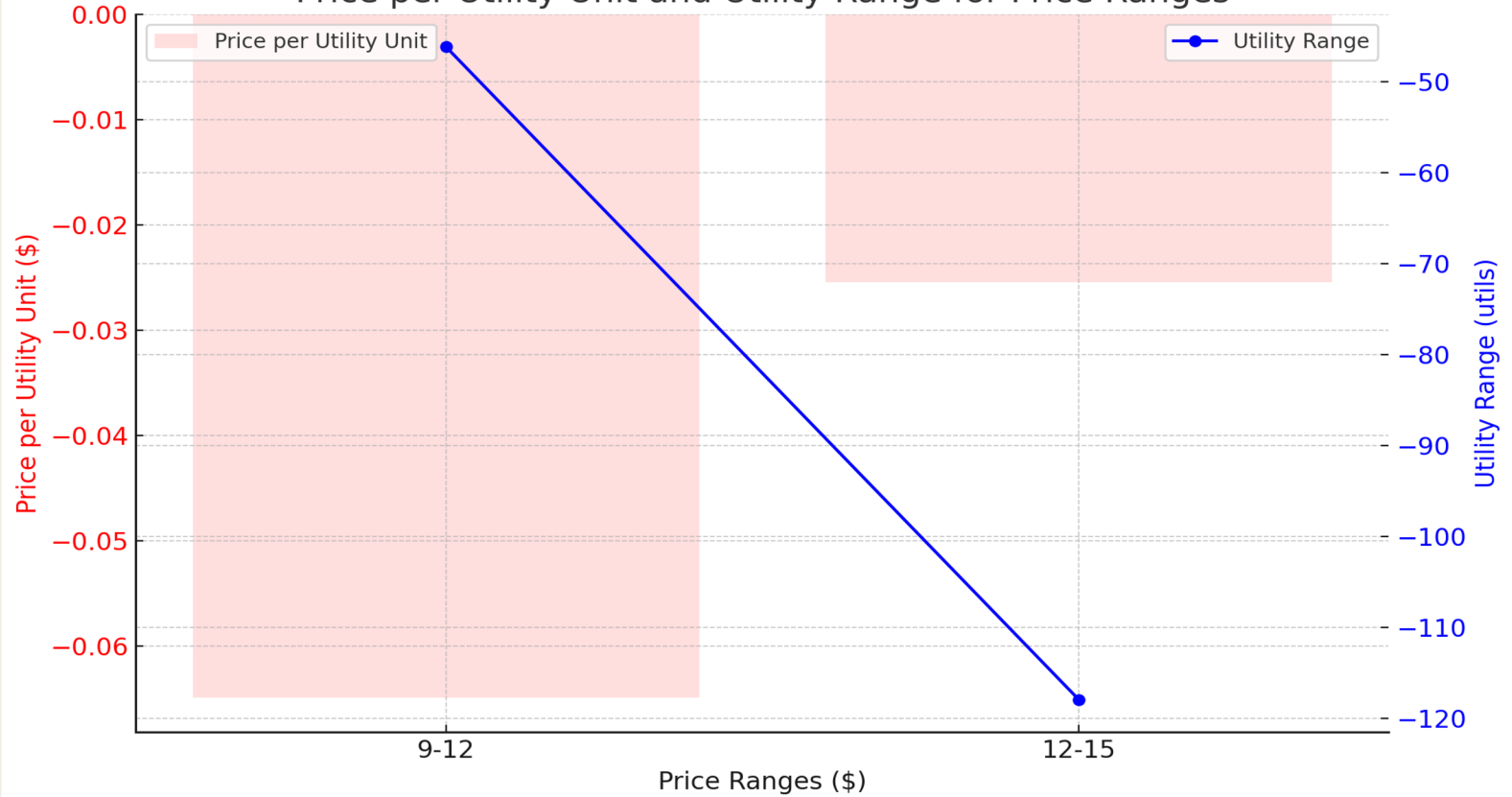
Thank You!

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Appendix

Price per Utility Unit and Utility Range for Price Ranges



GRADS	Total	1-3 meals a week	3+ meals a week	never
Survey	91	57.00	12.00	22.00
Sample Proportion		0.63	0.13	0.24
Population	3200	2004.40	421.98	773.63
Avg Meals		2.00	4.00	0.00
Total meals	5696.7	4008.79	1687.91	0.00

UNDERGRADS	Total	1-3 meals a week	3+ meals a week	never
Survey	10	6.00	2.00	2.00
Sample Proportion		0.60	0.20	0.20
Population	3200	1920.00	640.00	640.00
Avg Meals		2.00	4.00	0.00
Total meals	6400	3840.00	2560.00	0.00

Faculty/Staff	Total	1-3 meals a week	3+ meals a week	never
Survey	10	7.00	2.00	1.00
Sample Proportion		0.70	0.20	0.10
Population	400	280.00	80.00	40.00
Avg Meals		2.00	4.00	0.00
Total meals	880	560.00	320.00	0.00

Computation for WTP

Utility Range for Price

Utility for \$9: 70.11

Utility for \$12: 23.91

Utility for \$15: -94.02

Utility Range = $U(\$9) - U(\$15) =$

$70.11 - (-94.02) = 164.13$ utils

Price Range = $\$15 - \$9 = 6$ dollars

Price per Utility Unit =

Price Range / Utility Range =

$6 / 164.13 = 0.03655$ dollars per util

0.02789 dollars per util

1. Fresh vs. Frozen

Utility difference = $U(\text{Fresh}) - U(\text{Frozen}) = 53.01 - (-53.01) = 106.02$ utils

WTP = Utility difference * Price per Utility Unit

WTP = $106.02 \times 0.03655 = 3.87$ dollars

Customers are willing to pay \$3.87 more for Fresh over Frozen.

2. Vegan/Vegetarian Dish (Yes vs. No)

Utility difference = $U(\text{Yes}) - U(\text{No}) = 0.35 - (-0.35) = 0.70$ utils

WTP = Utility difference * Price per Utility Unit

WTP = $0.70 \times 0.03655 = 0.03$ dollars

Customers are willing to pay \$0.03 more for a Vegan/Vegetarian Dish.

3. Rotating Menu (Yes vs. No)

Utility difference = $U(\text{Yes}) - U(\text{No}) = 26.07 - (-26.07) = 52.14$ utils

WTP = Utility difference * Price per Utility Unit

WTP = $52.14 \times 0.03655 = 1.91$ dollars

Customers are willing to pay \$1.91 more for a Rotating Menu.

4. Portion Size (Large vs. Medium)

Utility difference = $U(\text{Large}) - U(\text{Medium}) = 2.25 - (-2.25) = 4.50$ utils

WTP = Utility difference * Price per Utility Unit

WTP = $4.50 \times 0.03655 = 0.16$ dollars

Customers are willing to pay \$0.16 more for a Large Portion over Medium.

Cost of Goods Sold (COGS):

- Fresh food typically costs \$6 per unit.
- COGS includes the cost of ingredients, preparation, and packaging.

Operational Costs:

- **Electricity:** \$10–\$20 per month (varies by machine type and usage).
- **Machine Maintenance:** \$50–\$100 per month.
- **Restocking Labor:** If outsourced, around \$15–\$20 per hour.
- **Waste/Spoilage:** Expect spoilage to account for 5–15% of inventory.
 - Example: At \$6 per unit, spoilage of 10% for 1,000 items = \$600/month.
- **Leasing Costs:** If the machine is leased, costs range from \$100–\$200 per month.

Marketing and Miscellaneous:

- Flyers, promotions, or digital displays could add \$50–\$100 per month.